

SHRIKRISHNA BHAT K MSc, PhD (Statistics)

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PROFESSIONAL SUMMARY

Statistician (PhD, Pondicherry University, awarded March 2026) with 6+ years of applied experience in survey analytics, statistical modelling, and reproducible data science. Developed probabilistic distance-based clustering methods and authored an R package on CRAN; peer-reviewed publications and conference presentations. Experienced in design-consistent survey estimation (weighting, non-response adjustment, post-stratification), logistic and regression modelling, and composite-index construction for large household surveys. Practical experience delivering policy-relevant analyses for government and research partners (MeitY — GoI; Government of Tamil Nadu; Survey Research Center, University of Michigan). Four semesters of teaching experience (Linear Models, Design of Experiments, R programming) and fluent in R, Python, and Stata for end-to-end reproducible pipelines.

EDUCATION

PhD in Statistics

Research Area: Cluster Analysis, Block Clustering, and Cluster diagnostics

Pondicherry University

Feb 2021 – Mar 2026

- Research Supervisor: Dr. Kiruthika
- Thesis Title: *Contributions to Block Clustering and its Diagnostic Measures*
- Developed novel **probabilistic distance-based clustering algorithms** (machine learning), implementing and benchmarking them as open-source R packages; methods validated on real-world socio-economic and biological datasets.
- **Teaching Assistant** to research supervisor: taught *Linear Models* and *Design of Experiments* using R for 4 semesters, and *R Programming* for 2 semesters at the Department of Statistics, Pondicherry University.

Master of Science in Statistics

CGPA: 7.75/10

Pondicherry University

Jul 2016 – Apr 2018

Bachelor of Science in Mathematics, Statistics, and Physics

Percentage: 80.31%

Andhra Loyola College

Jul 2013 – Apr 2016

EXPERIENCE

Madras Institute of Development Studies

Chennai

Data Consultant - Bridging the Gap in Digital Economy: Enabling Drivers and Constraining Factors Impeding Digital Inclusivity

Dec 1, 2025 – Mar 31, 2026

- Designed the **GAHDDI**: a three-pillar household-level composite index (digital access, capability, financial participation) with a $1 - |\text{gap}|$ **gender-equality weight** that penalises gaps between male and female members in the same household.
- Built Stata pipelines integrating household, roster, and respondent modules for **4,199 households across four aspirational districts** (Assam, Odisha, Rajasthan, Tamil Nadu); produced pillar and overall scores by district and rural/urban sector.
- Mapped the index to the **NSS 80th Round CSMT (2025)** for state- and national-level scaling, and authored the **technical report for MeitY, Government of India**.

Cushman and Wakefield

Chennai

Senior Statistician

May 7, 2025 – Jun 30, 2025

- Lead statistician for the **Housing Demand and Supply Study** for the Chennai Third Master Plan 2046, commissioned by CMDA.
- Built **regression demand models** on household survey data to identify determinants and quantify needs by location and income class.
- Produced **effective demand projections** for home purchase and social-housing rental for 2031, 2036, and 2046 by zone and affordability tier.

Centre for Social Justice and Equity, Madras School of Social Work

Chennai

Data Analyst - Socio Economic Survey of Scheduled Tribes in Tamil Nadu

Sep 7, 2024 – Dec 31, 2024

- Data analyst on the state-wide **Socio-Economic Survey of Scheduled Tribes in Tamil Nadu** (Government of Tamil Nadu).
- Implemented **real-time data quality monitoring** during active fieldwork, including outlier flags, enumerator-level error reports, and post-coding reviews.
- Produced summary tables and rural/urban progress reports at district, block, and panchayat levels.

Madras Institute of Development Studies

Chennai

Consultant - Tamil Nadu Household Panel Survey

Aug 23, 2022 – Aug 24, 2024

- Half-time consultant and **data analysis supervisor** for TNHPS, a longitudinal study covering **20,000+ households**.
- Owned the STATA-based data monitoring, cleaning, and analysis workflow; supervised a team of analysts and field statisticians across waves.

Statistical Data Analyst - Tamil Nadu Household Panel Survey

Feb 21, 2019 – Mar 31, 2021

- Analysed the Pre-Baseline phase of TNHPS, a longitudinal survey covering **240,000+ households** across Tamil Nadu.
- Built **logistic non-response adjustment models, calibration-based survey weights**, and post-stratification corrections to produce design-consistent estimates aligned with Census benchmarks.
- Major contributor to the **Tamil Nadu Covid Pulse Survey (TNCPS)**: cleaned and analysed 12,000+ households across three waves and quantified pandemic-driven income and employment shocks using difference-in-differences models.
- **Partners**: Department of Economics & Statistics, Government of Tamil Nadu; Survey Research Center, University of Michigan.

Core CarbonX Solutions Pvt. Ltd.

Hyderabad

Associate Consultant

Jun 10, 2018 – Feb 15, 2019

- Developed a **Multiple Criteria Decision Making** model in R to prioritise Industrial Parks in Telangana for TSIIC Ltd.
- Coordinated two Swachh Bharat Mission Solid Waste Management workshops in Tirupati (funded by NIUA).

SKILLS AND LANGUAGES

Programming Languages	Python · R
Machine Learning	scikit-learn · Logistic Regression · Classification · Clustering · Ensemble Methods
Statistical Tools	STATA · SPSS · Minitab
Data Tools	MS Excel · SQL · Power BI · Tableau 
Version Control	GitHub
Other Tools	L ^A T _E X · SurveyCTO · MS Office
Languages Known	English (Fluent) · Kannada (Native) · Telugu (Native) · Hindi (Fluent) · Tulu (Native) · Tamil (Basic)

PUBLICATIONS

1. Kripa Ananth Pur and **Shrikrishna Bhat K** (Mar. 2026). *Gender-Adjusted Household Digital Divide Index (GAHDDI): A Household-Level Index of Digital Divide with Gender Adjustments*. Technical Report. Madras Institute of Development Studies (MIDS), Chennai; Ministry of Electronics and Information Technology, Government of India.
2. **Shrikrishna Bhat K** and Kiruthika C (June 25, 2025a). *Block Probabilistic Distance Clustering: A Unified Framework and Evaluation*. Preprint, Version 1. DOI: 10.21203/rs.3.rs-6973596/v1. ResearchSquare: rs-6973596/v1.
3. **Shrikrishna Bhat K** and Kiruthika C (2024). “Some density-based silhouette diagnostics for soft clustering algorithms”. In: *Communications in Statistics: Case Studies, Data Analysis and Applications* 10(3–4), pp. 221–238. DOI: 10.1080/23737484.2024.2408534.

PACKAGES

1. **Shrikrishna Bhat K** and Kiruthika C (2025b). *blockclusterPDQ: An R Package for Block Probabilistic Distance Clustering*. GitHub: kskbhat/blockclusterPDQ.
2. **Shrikrishna Bhat K** and Kiruthika C (2025c). *Silhouette: Proximity Measure Based Diagnostics for Standard, Soft, and Multi-Way Clustering*. R package version 0.9.6. DOI: 10.32614/CRAN.package.Silhouette. CRAN: Silhouette. URL: <https://kskbhat.github.io/Silhouette>.

HONORS AND AWARDS

- Certificate of Appreciation, Directorate of Tribal Welfare, Government of Tamil Nadu — 29 July 2025. For data cleaning and analysis contributions to the Socio-Economic Survey of Scheduled Tribes in Tamil Nadu (August–December 2024).

WORKSHOPS AND CONFERENCES

Papers Presented

1. “Probabilistic Distance Coclustering for Ordinal Data” (Feb. 6–8, 2024). In: *International Conference on Recent Advances of Probability and Statistics in Interdisciplinary Research (RAPSIR-2024) in conjunction with the 43rd Annual Convention of ISPS*. Paper presented. University of Allahabad, India.
2. “Density-Based Silhouettes to Evaluate the Performance of Soft Clustering Algorithms” (Sept. 1–2, 2023). In: *International Conference on Statistical Theory and its Applications (ICSTA-2023)*. Virtual paper presentation. Bharathiar University, India.
3. “Probability Density-Based Clustering” (Dec. 30, 2022). In: *Annual Conference of International Indian Statistical Association (IISA 2022)*. Paper presented. IISc, Bengaluru, India.

Posters Presented

1. “Exploring Block Clustering with Probabilistic Distance: Theory and Validation” (Dec. 27–31, 2024). In: *Annual Conference of International Indian Statistical Association (IISA 2024)*. Student Poster Competition. CUSAT, Cochin, India.

Participated

1. *Pre-Annual Convention Workshop on Advanced Data Science Techniques* (Feb. 5, 2024). In conjunction with the 43rd Annual Convention of ISPS. Participation. University of Allahabad, India.
2. *24th Annual Conference of SSCA (RASTA-2022)* (Feb. 23–27, 2022). Online event on Recent Advances in Statistical Theory and Applications, ICAR-NAARM, Hyderabad. Participation. Hyderabad, India.
3. *International Conference on Advances in Statistical Methods and Applications (ICASMA-2022)* (Jan. 24–25, 2022). Organised by the Department of Statistics, University of Madras. Participation. Chennai, India.
4. *28th International Workshop on Matrices and Statistics (IWMS 2021)* (Dec. 13–15, 2021). Hosted by Centre for Advanced Research in Applied Mathematics and Statistics, MAHE, Manipal. Participation. Manipal, India.
5. *International Workshop on Data Science (DATUM 2021)* (Sept. 18–20, 2021). In association with the International Indian Statistical Association, DA-IICT, Gandhinagar. Participation. Gandhinagar, India.